

Figure 9.2 Rickets

9.2.2. Phosphorus

Phosphorus is the principal anion of the cell. It is widely present as phosphates in proteins, nucleic acids and other various cellular components.Phosphorus and calcium are related in sources and metabolism.

Functions

1. Phosphorus is essential for formation of bones and teeth.

2. High-energy phosphate compounds like ATP (Adenosine Tri Phosphate) and Creatine Phosphate play a role in storage and transport of energy.

3. Phosphorylation and dephosphorylation reactions modify the activity of many enzymes.

4. Phospholipids, the important constituent lipid of cell membranes and nervous tissue contain phosphorus.

5. Acid-base balance is maintained by phosphate buffer in the kidneys.

6. Several co-enzymes such as $NADP^+$ and TPP involved in enzymatic reactions contain phosphate.

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